Aline de Matos Valerio, Ph.D.

Postdoctoral Associate, O'Neill School of Public and Environmental Affairs, Indiana University Bloomington
alinedmv@gmail.com | ademato@iu.edu
Google Scholar: scholar.google.com/citations?user=yqwg1e8AAAAJ | ORCID: 0000-0002-4742-771X

EDUCATION

2018 **Ph.D., Remote Sensing**, National Institute for Space Research (INPE), Brazil. Dissertation: "Assessment of the spatiotemporal variability of optical and biogeochemical parameters in the lower Amazon region and of the carbon content in the Amazon River Continuum using in situ and remote sensing data."

2009 **M.Sc., Remote Sensing**, National Institute for Space Research (INPE), Brazil. Thesis: "Using remote sensing and in situ data to assess the spectral behavior of the Manso Reservoir, MT, Brazil."

2008 **Specialization, Marine Biology**, Taubaté University (UNITAU), Brazil.

2005 **B.S., Biology**, Taubaté University (UNITAU), Brazil.

PROFESSIONAL APPOINTMENTS

2025–present **Postdoctoral Associate**, O'Neill School of Public and Environmental Affairs, Indiana University Bloomington, USA. Environmental Change & Sustainability Lab (PI: R.M. Almeida).

2020–2024 **Postdoctoral Fellow (FAPESP)**, University of São Paulo (USP), Brazil.

2019–2020 **Postdoctoral Fellow (PNPD/CAPES)**, National Institute for Space Research (INPE), Brazil.

2010–2013 **Research Fellow (DTI-PCI and FUNCATE)** at the National Institute for Space Research (INPE), Brazil.

COMPETITIVE FELLOWSHIPS (as PI)

2020–2024 Postdoctoral Fellowship, São Paulo Research Foundation (FAPESP), University of São Paulo. "Spatio-temporal variability of pCO_2 in the Amazon River by remote sensing of water color: from the River to the Plume." R$492,000 total.

2019–2020 Postdoctoral Fellowship, National Post-Doctoral Program (PNPD/CAPES), INPE. "Spatio-temporal variability of pCO_2 estimated by remote sensing at the mouth of the Amazon River." R$60,000.

2013–2018 Ph.D. Fellowship, CAPES, INPE. R$125,400.

2010–2013 Research fellowships, DTI-PCI (CNPq) and FUNCATE, INPE: Doce and Paraíba do Sul river plumes; regional characterization of strategic marine areas; Campos Bay. R$125,200 combined.

2007–2009 M.Sc. Fellowship, National Council for Scientific and Technological Development (CNPq), INPE. R$28,800.

PUBLICATIONS

Google Scholar (October 2026): 1,065 citations; h-index 10; i10-index 10.

Peer-reviewed journal articles

17. Valerio, A.M.; Kampel, M.; Vantrepotte, V.; Ward, N.D.; Almeida, R.M.; Richey, J.E. (2026). Remotely sensed coloured dissolved organic matter distribution in the Amazon River Continuum. **International Journal of Remote Sensing**, 1–43. https://doi.org/10.1080/01431161.2026.2720314

16. Pacheco, F.; Heilpern, S.A.; Almeida, R.M.; Sethi, S.; … Gomes, C.; Valerio, A.M.; Flecker, A.S. (2026). Land use transitions and carbon emissions of aquaculture expansion in the Brazilian Amazon. **Science Advances**, 12, eaec5472. https://doi.org/10.1126/sciadv.aec5472

15. Valerio, A.M.; Montanari, J.L.; Kampel, M.; Ward, N.D.; Richey, J.E.; Cunha, A.C. (2025). Chlorophyll-a dynamics in the Lower Amazon River: insights from in situ and hyperspectral remote sensing using OCI-PACE. **International Journal of Remote Sensing**, 1–26. https://doi.org/10.1080/01431161.2025.2596811

14. Maciel, D.A.; Barbosa, C.C.F.; de Moraes Novo, E.M.L.; … Valerio, A.M.; et al. (2025). A bio-optical database for the remote sensing of water quality in BRAZil coAstal and inland waters (BRAZA). **Scientific Data**, 12, 1270. https://doi.org/10.1038/s41597-025-05609-1

13. Valerio, A.M.; Kampel, M.; Vantrepotte, V.; Ballester, V.; Richey, J.E. (2024). Assessment of atmospheric correction algorithms for Sentinel-3 OLCI in the Amazon River Continuum. **Remote Sensing**, 16(14), 2663. https://doi.org/10.3390/rs16142663

12. Curra-Sánchez, E.D.; Valerio, A.M.; Lara, C.; García-Tuñon, W.; Broitman, B.R.; Saldías, G.S.; … Vargas, C.A. (2024). CDOM dynamics in two coastal zones influenced by contrasting land uses in northern Patagonia. **Estuarine, Coastal and Shelf Science**, 306, 108897. https://doi.org/10.1016/j.ecss.2024.108897

11. Moreira, D.L.; Dalto, A.G.; Figueiredo, A.G.; Valerio, A.M.; Detoni, A.M.S.; Bonecker, A.C.T.; … Sousa, S.H.M. (2023). Multidisciplinary scientific cruises for environmental characterization in the Santos Basin: methods and sampling design. **Ocean and Coastal Research**, 71(suppl. 3), e23022. https://doi.org/10.1590/2675-2824071.22072dlm

10. Cesar, G.M.; Rudorff, N.M.; Oliveira, A.L.; Valerio, A.M.; Chuqui, M.G.; Pompeu, M.; … Kampel, M. (2023). Bio-optical properties of South Brazil Bight coastal waters and implications for satellite chlorophyll-a concentration retrieval. **International Journal of Remote Sensing**, 44(7), 2428–2457. https://doi.org/10.1080/01431161.2023.2201385

9. Valerio, A.M.; Kampel, M.; Ward, N.D.; Sawakuchi, H.O.; Cunha, A.C.; Richey, J.E. (2021). CO_2 partial pressure and fluxes in the Amazon River plume using in situ and remote sensing data. **Continental Shelf Research**, 215, 104348. https://doi.org/10.1016/j.csr.2021.104348

8. Valerio, A.M.; Kampel, M.; Vantrepotte, V.; Ward, N.D.; Richey, J.E. (2021). Optical classification of lower Amazon waters based on in situ data and Sentinel-3 Ocean and Land Color Instrument imagery. **Remote Sensing**, 13(16), 3057. https://doi.org/10.3390/rs13163057

7. Kuhn, C.; Valerio, A.M.; Ward, N.; Loken, L.; Sawakuchi, H.O.; Kampel, M.; … Butman, D. (2019). Performance of Landsat-8 and Sentinel-2 surface reflectance products for river remote sensing retrievals of chlorophyll-a and turbidity. **Remote Sensing of Environment**, 224, 104–118. https://doi.org/10.1016/j.rse.2019.01.023

6. Valerio, A.M.; Kampel, M.; Vantrepotte, V.; Ward, N.D.; Sawakuchi, H.O.; Less, D.F.D.S.; … Richey, J.E. (2018). Using CDOM optical properties for estimating DOC concentrations and pCO_2 in the Lower Amazon River. **Optics Express**, 26(14), A657–A677. https://doi.org/10.1364/oe.26.00a657

5. Ward, N.D.; Sawakuchi, H.O.; Neu, V.; Less, D.F.S.; Valerio, A.M.; Cunha, A.C.; … Keil, R.G. (2018). Velocity-amplified microbial respiration rates in the lower Amazon River. **Limnology and Oceanography Letters**, 3(3), 265–274. https://doi.org/10.1002/lol2.10062

4. Sawakuchi, H.O.; Neu, V.; Ward, N.D.; Barros, M.L.C.; Valerio, A.M.; Gagne-Maynard, W.; … Richey, J.E. (2017). Carbon dioxide emissions along the lower Amazon River. **Frontiers in Marine Science**, 4, 76. https://doi.org/10.3389/fmars.2017.00076

3. Gagne-Maynard, W.C.; Ward, N.D.; Keil, R.G.; Sawakuchi, H.O.; Da Cunha, A.C.; Neu, V.; Brito, D.C.; Less, D.F.D.S.; Diniz, J.E.M.; Valerio, A.M.; Kampel, M.; Krusche, A.V.; Richey, J.E. (2017). Evaluation of primary production in the lower Amazon River based on a dissolved oxygen stable isotopic mass balance. **Frontiers in Marine Science**, 4, 26. https://doi.org/10.3389/fmars.2017.00026
2. Ward, N.D.; Bianchi, T.S.; Sawakuchi, H.O.; Gagne-Maynard, W.; Cunha, A.C.; Brito, D.C.; Neu, V.; Valerio, A.M.; Silva, R.; Krusche, A.V.; Richey, J.E.; Keil, R.G. (2016). The reactivity of plant-derived organic matter and the potential importance of priming effects along the lower Amazon River. **Journal of Geophysical Research: Biogeosciences**, 121(6), 1522–1539. https://doi.org/10.1002/2016JG003342
1. Valerio, A.M.; Kampel, M.; Assireu, A.T.; Stech, J.L. (2012). The asymmetric fragmentation operator applied to meteo-limnological time series in a tropical reservoir. **Ecological Informatics**, 8, 29–36. https://doi.org/10.1016/j.ecoinf.2011.12.003

Submitted

1. Almeida, R.M.; Arantes, C.; Bhattarai, S.; Cardoso, S.; Chowdhury, A.F.M.K.; Del Claro, A.; Doria, C.; Fluet-Chouinard, E.; … Valerio, A.M.; Wild, T.; Wilcox, A.; Gomes, C.; Flecker, A. Freezing hydropower's socio-environmental footprint on rivers. Submitted to **Nature**.

In preparation

3. Valerio, A.M.; Reji, F.; Pacheco, F.; Nair, S.; Amin, H.; Nobre, R.; Cardoso, S.J.; Flecker, A.S.; Gomes, C.P.; Almeida, R.M. A global geospatial dataset of aquatic photovoltaic installations and host waterbodies. For **Scientific Data**.
2. Valerio, A.M.; Fluet-Chouinard, E.; Nobre, R.; Cardoso, S.J.; Almeida, R.M. Thermal responses of lakes and reservoirs to floating solar photovoltaics. For **Nature Water**.
1. Valerio, A.M.; Fluet-Chouinard, E.; Cardoso, S.J.; Almeida, R.M. Global chlorophyll-a responses to floating photovoltaic installations. For **Nature Water**.

Book chapters

2. Kampel, M.; Oliveira, N.R.; Cortivo, F.D.; Santos, J.F.C.; Freitas, L.B.; Cesar, G.M.; Valerio, A.M. (2019). Caracterização sazonal de campos superficiais estimados por sensoriamento remoto na bacia sedimentar de Sergipe-Alagoas. In: Moreira, D.L. et al. (Eds.), *Ambiente Pelágico da Bacia de Sergipe-Alagoas*. Editora UFS.
1. Valerio, A.M.; Stech, J.L.; Kampel, M. (2011). Sensoriamento remoto orbital e de superfície para o estudo do comportamento do corpo de água do reservatório de Manso, MT, Brasil. In: Alcântara, E.H.; Novo, E.M.L.M.; Stech, J.L. (Eds.), *Novas tecnologias para o monitoramento e estudo de reservatórios hidrelétricos e grandes lagos*. Editora Parêntese.

DATA AND SOFTWARE PRODUCTS

AqPV global inventory (2026). Geospatial database of 3,277 aquatic photovoltaic installations on 2,545 host water bodies in 36 countries, all delineated by A.M. Valerio, with installation dates (NSPI-1 index of Li et al. 2025, adapted to a panel-minus-water change-point method), structure and host water-body attributes (multimodal AI classification, blind-validated). Public release with the *Scientific Data* paper.

AqPV web tool (2026). Interactive web platform to explore aquatic PV installations and water-quality and thermal indices, built by F. Reji (M.S. student I co-mentored).

TEACHING EXPERIENCE

2026 Guest lecture, “Introduction to Remote Sensing: Understanding Our Planet from Space,” in SPEA-E 272 Introduction to Environmental Sciences (undergraduate), Indiana University Bloomington, April 20, 2026.

2015 Instructor, GIS applications in Biology (undergraduate, one semester), Universidade do Vale do Paraíba (UNIVAP), São José dos Campos, Brazil.

2024 Invited seminar, “Water Color Remote Sensing: Fundamentals and Applications for Aquatic Ecosystem Studies,” Universidad Católica de la Santísima Concepción (UCSC), Chile.

MENTORING

2026 Freny Reji, Sakshi Nair, Heeya Mineshkumar Amin, M.S. students, Luddy School of Informatics, Computing, and Engineering, Indiana University (co-mentor, Faculty Assistance in Data Science program). F. Reji built the AqPV web tool; S. Nair and H. Amin integrated the source datasets; all three are co-authors of the AqPV inventory paper.

2024 João Luiz Montanari, M.Sc. student, Graduate Program in Tropical Biodiversity, Federal University of Amapá, Brazil (co-supervisor and thesis committee member). Co-author, *International Journal of Remote Sensing* 2025.

INVITED TALKS AND PRESENTATIONS

Invited talks

2025 “Remote sensing carbon fraction estimates in the Amazon River Continuum: advances and challenges.” XXI Brazilian Remote Sensing Symposium (SBSR), Amazon Hydrology session, Salvador, Brazil.

2024 “Water Color Remote Sensing: Fundamentals and Applications for Aquatic Ecosystem Studies.” UCSC, Concepción, Chile.

Selected conference presentations

2026 Global chlorophyll-a responses to floating photovoltaic installations. ASLO Aquatic Sciences Meeting, Montréal, Canada.

2025 Identifying activity status of Amazonian fishponds using remote sensing. XXI SBSR, Salvador, Brazil.

2025 Retrieving colored dissolved organic matter using S3-OLCI in the Amazon River Continuum. XXI SBSR, Salvador, Brazil.

2024 Brazilian optical water types from remote sensing reflectance data using a smartphone camera. XX Latin American Congress of Marine Sciences (COLACMAR), Itajaí, Brazil.

RESEARCH COLLABORATIONS

2023–2024 International collaborator, “Bio-optical gradients along the river-estuary-coastal ocean continuum of northern Patagonia,” Fondecyt Regular 1230420 (PI: C. Lara), UCSC, Chile.

“Carbon footprint and impacts of aquaculture expansion in the Amazon,” FAPESP 2022/10443-6.

“Diagnosis and environmental restoration strategies for the Paraíba do Norte River basin,” FAPESP 2022/08775-0.

"COCOBRAZ: Brazilian coastal water biogeochemical quality over the last two decades from satellite observation," FAPESP 2021/04128-8.

"Biogeochemical blind spots along the lower Amazon River Continuum: from land to the atmosphere and ocean," FAPESP 2018/18491-4.

PROFESSIONAL SERVICE

Ad hoc reviewer: Remote Sensing of Environment; Journal of Geophysical Research: Biogeosciences; Earth System Science Data; Journal of Hydrology; International Journal of Remote Sensing; Remote Sensing; Optics Express; Applied Optics; Earth and Space Science; Marine Pollution Bulletin; Journal of Cleaner Production; Journal of Marine Science and Engineering; Journal of Sustainability.

Thesis examination committees

2026	Ph.D., Ikram Salah Salah, Université du Littoral Côte d'Opale (ULCO), France.
2025	Ph.D., João Felipe Cardoso dos Santos, INPE, Brazil.
2025	M.Sc., Pedro Ferrini Manhães Bacellar, INPE, Brazil.
2022	Ph.D., Elizabeth Curra Sánchez, Universidad de Concepción, Chile.

LANGUAGES

Portuguese (native), English (fluent), Spanish (fluent).